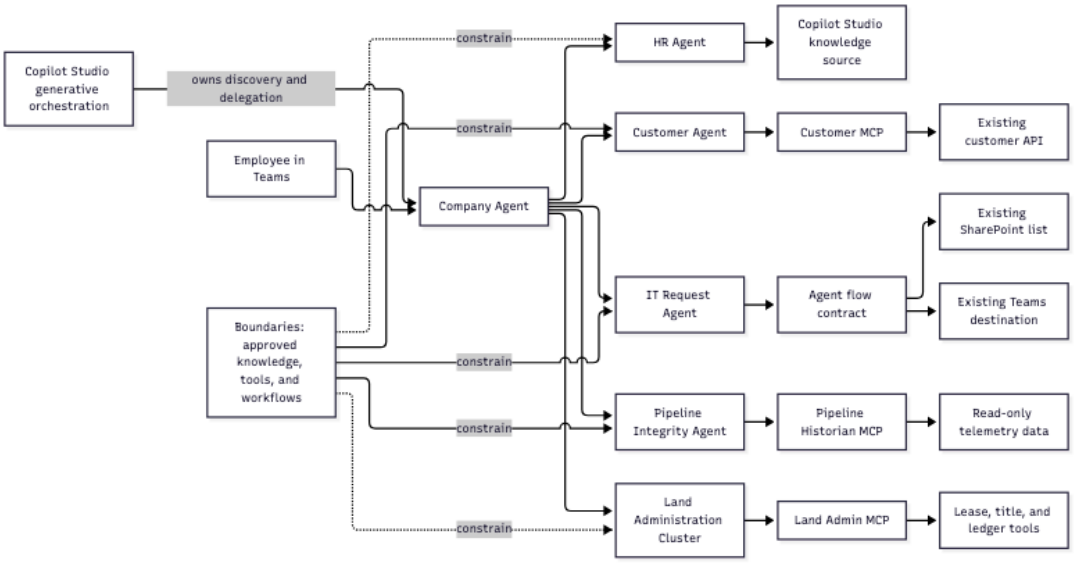
Company_Agent_Arch_Reference

Company Agent architecture reference

This repository is a small reference solution for composing bounded Copilot Studio agents around existing applications and deterministic workflows.



Composition boundary

- **Company Agent** is the employee-facing front door. Copilot Studio owns agent discovery, delegation, and conversation orchestration.
- **HR Agent** answers general policy questions from approved Copilot Studio knowledge. It does not access employee records or make employment decisions.
- **Customer Agent** uses the read-only Customer MCP server. The MCP server calls the existing customer API and reports only the fields that API returns.
- **IT Request Agent** gathers `employeeName`, `requestType`, `description`, and `isUrgent`, then calls the approved Agent flow. The flow reuses the existing SharePoint list and Teams destination.
- **Pipeline Integrity Agent** uses the Pipeline Historian MCP server for read-only telemetry and sequential compliance screening.
- **Land Administration Cluster** uses the Land Admin MCP server for lease extraction, title review, and confirmed ledger posting.

Repository contracts

The repository is the source of truth for local agent and workflow contracts.

Boundary	Contract or implementation	Result
Front door	`agents/company-agent/agent.yaml`	Native Copilot Studio delegation
HR	`agents/hr-agent/agent.yaml`	Approved policy guidance
Customer	`agents/customer-agent/agent.yaml`, `customer_mcp/server.py`	Read-only customer lookup
IT request	`agents/it-request-agent/agent.yaml`, `it-request-workflow/agent-flow.contract.json`	Service Request ID and status
Pipeline	`agents/pipeline-integrity/agent.yaml`, `pipeline_historian_mcp/`	Observed telemetry and screening result
Land administration	`agents/land-admin/agent.yaml`, `land_admin_mcp/`	Extracted facts and confirmed ledger result

Workflow boundary

The IT Agent flow is an agent-callable entry point to the existing business process. It does not replace the Forms-triggered flow.

```
Agent inputs
  -> normalize and validate
  -> SharePoint Create item in the existing list
  -> SharePoint Update item with SR-###
  -> Teams Post card to the existing destination
  -> serviceRequestId, status, and result
```

The flow must return the observed Service Request ID and status before the agent reports that the request succeeded. The flow must not create a new Forms flow, SharePoint list, Teams team, or replacement workflow.

Data and authority boundaries

- Approved HR knowledge is authoritative for general policy guidance.
- The existing customer API is authoritative for customer lookup results.
- MCP tools expose narrow, read-only or explicitly confirmed operations.
- The existing SharePoint and Teams process remains authoritative for IT request creation and notification.

- Pipeline and land-admin results remain tool-backed observations. Compliance, title, ownership, and approval decisions remain with qualified humans.
- PAC workspaces and tenant deployment artifacts are local outputs. They are not repository source files.

What this reference does not add

This architecture does not add a repository-side router, a central orchestration API, a generalized department framework, or unrestricted database access. It keeps discovery and delegation in Copilot Studio and keeps deterministic workflows and existing APIs authoritative.